

Ravindran S

ravindrans.dev@gmail.com · +91 82487 02358 · Chennai, India

linkedin.com/in/ravindran-s-982702327 · github.com/ravindran-dev · leetcode.com/u/RAVINDRAN_S

PROFESSIONAL SUMMARY

Computer Science undergraduate (AI & ML) with hands-on experience in edge inference, computer vision, and full-stack systems engineering. Built sub-1-GFLOP real-time object detection models for drone hardware and won 2 national hackathons. LeetCode Knight (2085 rating), ICPC 2025 participant. Proficient across AOSP, PyTorch, TensorRT, Go, Rust, CUDA, and distributed backend systems.

EDUCATION

Chennai Institute of Technology — B.E. Computer Science & Engineering (AI & ML)

Sept 2024 – Present

Cumulative GPA: 9.24 / 10

Chennai, India

PROFESSIONAL EXPERIENCE

Drone Technology Team — CIT (NIDAR Competition)

Oct 2025 – Dec 2025

Object Detection in Drones

Chennai, India

Engineered a sub-1 GFLOP real-time object detection model (NanoDet backbone, YOLO-style heads), achieving **96% mAP** and **30+ FPS** on-device inference via **TensorRT INT8** quantization.

- Evaluated FP32 vs. INT8 and backbone trade-offs to meet strict compute constraints for autonomous drone payloads; presented results at **NIDAR 2025**.

AI/ML Virtual Internship — AICTE & EduSkills

Apr 2025 – Jun 2025

Full-Stack Development

Virtual

- Built and deployed a full-stack web application (**React** frontend, **Flask/Python** backend) serving ML model predictions via REST APIs, supporting **100+ concurrent** user requests during testing.

Open Source Contributor — AWS s2n-tls

- Contributed patches to **AWS s2n-tls** (C, TLS 1.3 implementation), addressing documentation gaps and correctness issues reviewed and merged by maintainers.

PROJECTS

MicroDet - Real-Time Edge Object Detection | TinyGrad, PyTorch, DeepStream SDK, TensorRT

- Architected a sub-1 GFLOP detection model (NanoDet backbone + YOLO decoupled heads); achieved **96% mAP@0.5** via TensorRT INT8 quantization, outperforming a MobileNetV2-SSD baseline (~89%) at equivalent FLOP budget.
- Presented at NIDAR 2025.

Archon - Security-Hardened AUR Package Manager | Rust, eBPF (Aya), Bubblewrap, libalpm

- Engineered a security-hardened AUR package manager for Arch Linux that resolves cross-source dependency graphs (official repos + AUR) and sandboxes untrusted build scripts, detecting malicious PKGBUILD behavior via **eBPF-based syscall monitoring** (execve, openat, connect) with **<5% build-time overhead**.

Machine Guard - IoT Anomaly Monitoring Pipeline | Kotlin, Python, MQTT, Firebase, IoT Sensors

- Architected an end-to-end IoT monitoring pipeline (sensor nodes → MQTT → Firebase → React dashboard) with **<200ms alert latency**; deployed an on-device quantized TFLite anomaly model at **94% accuracy**. Official submission @ Ti Forge 2026.

TECHNICAL SKILLS

- Languages:** C, C++, Python, JavaScript, TypeScript, SQL, Go, Rust, CUDA
- ML / AI:** PyTorch, TensorFlow Lite, TensorRT, TinyGrad, DeepStream SDK, LangChain, Sentence Transformers
- Systems & Infrastructure:** Linux, Docker, Nginx, Redis, MQTT, WebSockets, Git
- Web & Full-Stack:** React.js, FastAPI, Streamlit, Firebase
- Core Competencies:** AOSP, Computer Vision, Edge Inference, LLM Systems (RAG/Agents), Distributed Systems, GPU Optimization, Multithreading

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- LeetCode Peak Contest Rating: **2085** (Knight badge holder) · Codechef Peak Rating: **1716** (3★) · Codeforces Peak Rating: **1135** (Newbie)
- ICPC 2025 Preliminary Contest — **Global Rank 2605**
- Hackathon Wins (2):** Ti Forge Hackathon (Rs. 50,000 cash prize), AI for Sustainability Hackathon (\$125 + Amazon vouchers worth 300 AED)
- Finalist — TGF 2.0 TechSprint Hackathon (Top 10 of 300 teams)
- Open Source:** AWS s2n-tls contributor; SWOC participant

CERTIFICATIONS

- Cisco Networking Academy:** Python Fundamentals, Cybersecurity, Modern AI, Data Science, CCNA (90–100%)
- NPTEL (IIT Kharagpur):** Introduction to Internet of Things — Elite, **75%** (Jul–Oct 2025, 12-week course)
- NPTEL (IIT Roorkee):** Data Analytics with Python — Elite, **70%** (Jan–Apr 2026, 12-week course)
- MongoDB:** RAG with MongoDB, AI Data Strategy with MongoDB, Vector Search Fundamentals, AI Agents with MongoDB, MongoDB Basics for Students